

HBM Base Die:

How HBM Will Evolve Using Advanced Logic Processes

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Contents

- 1. New Opportunities of HBM Base-die**
- 2. Phase #1: xPU Area Reclamation**
- 3. Phase #2: Additional Functions**
- 4. Phase #3: 3D Integration**
- 5. Conclusion and Summary**

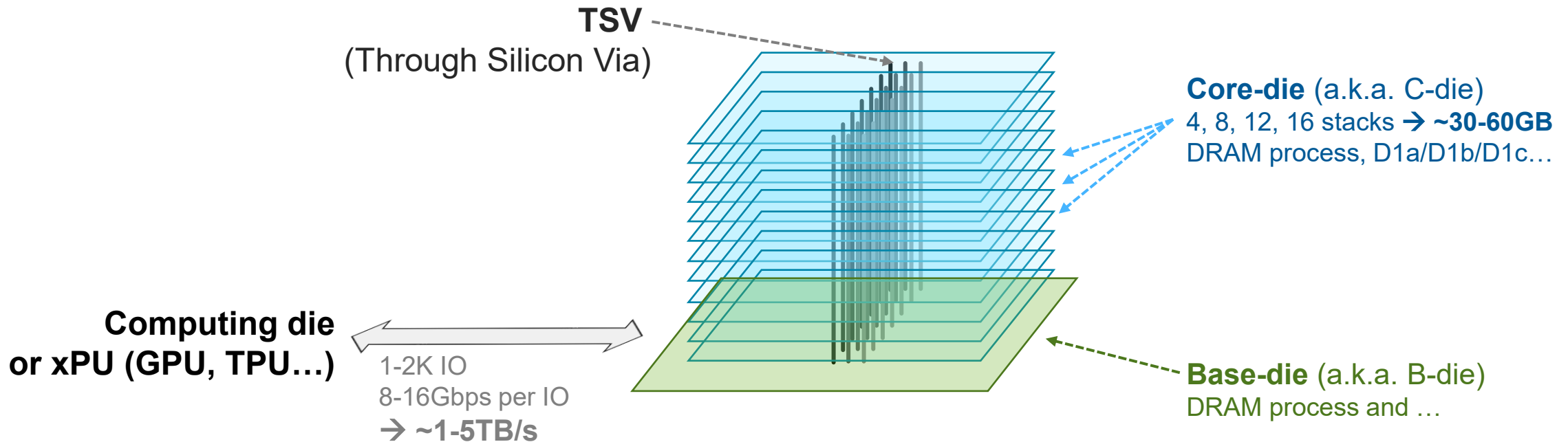
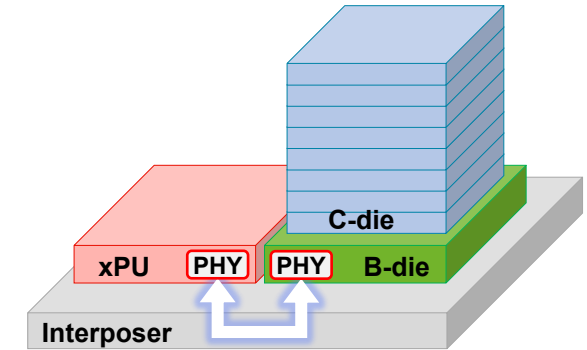
High Bandwidth Memory (HBM)

■ C-die role: DRAM cell & core

■ B-die role:

- Communication channel to the computing die (PHY-TSV)
- C-die test functionality and channels (DA, MBIST, IEEE1500, etc.)

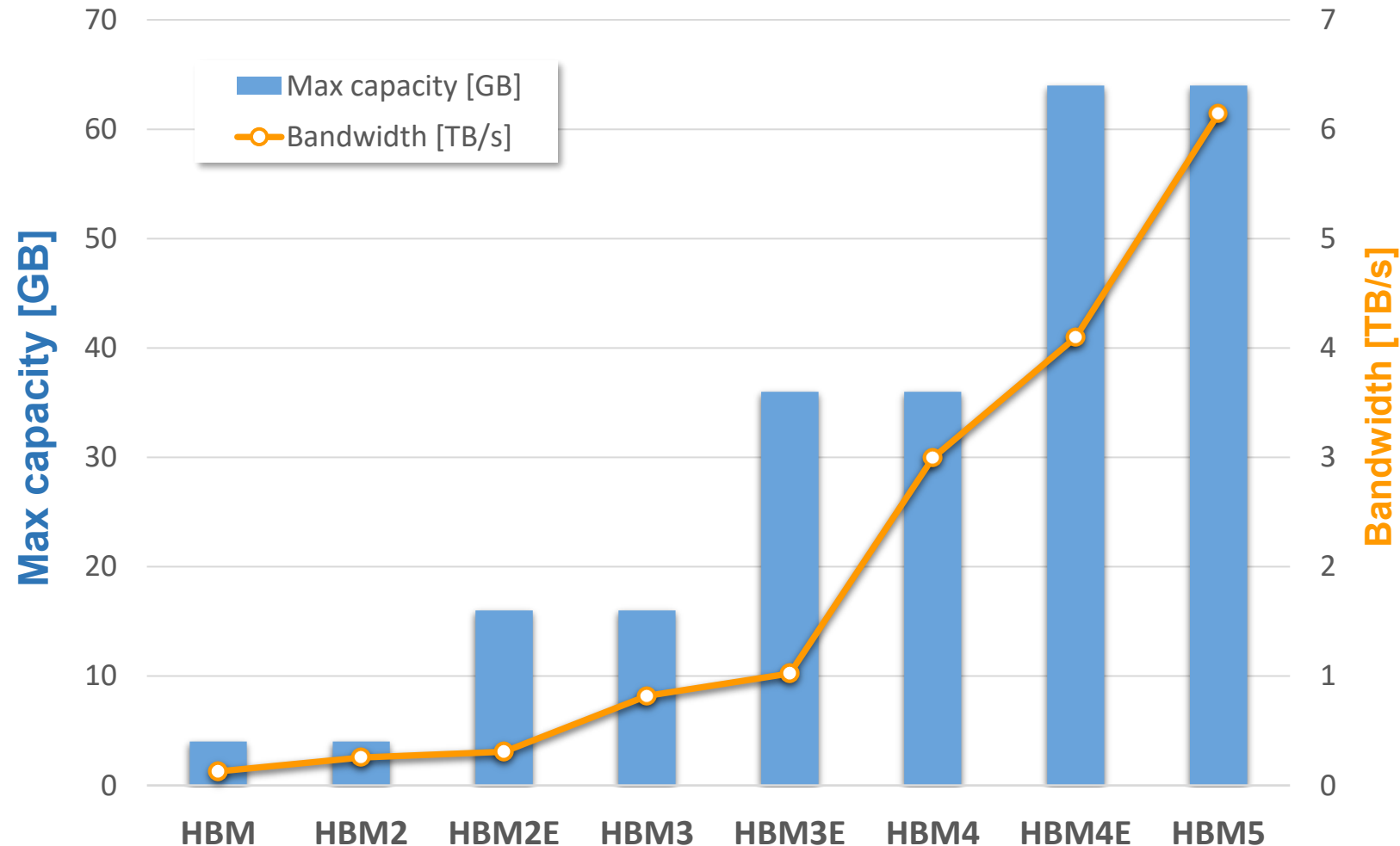
※ Direct Access



HBM Development History

■ Higher bandwidth: the primary driver of evolution

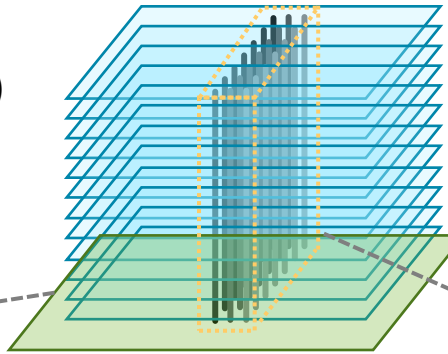
- Rapid bandwidth (BW) scaling requires fundamental changes in B-die.



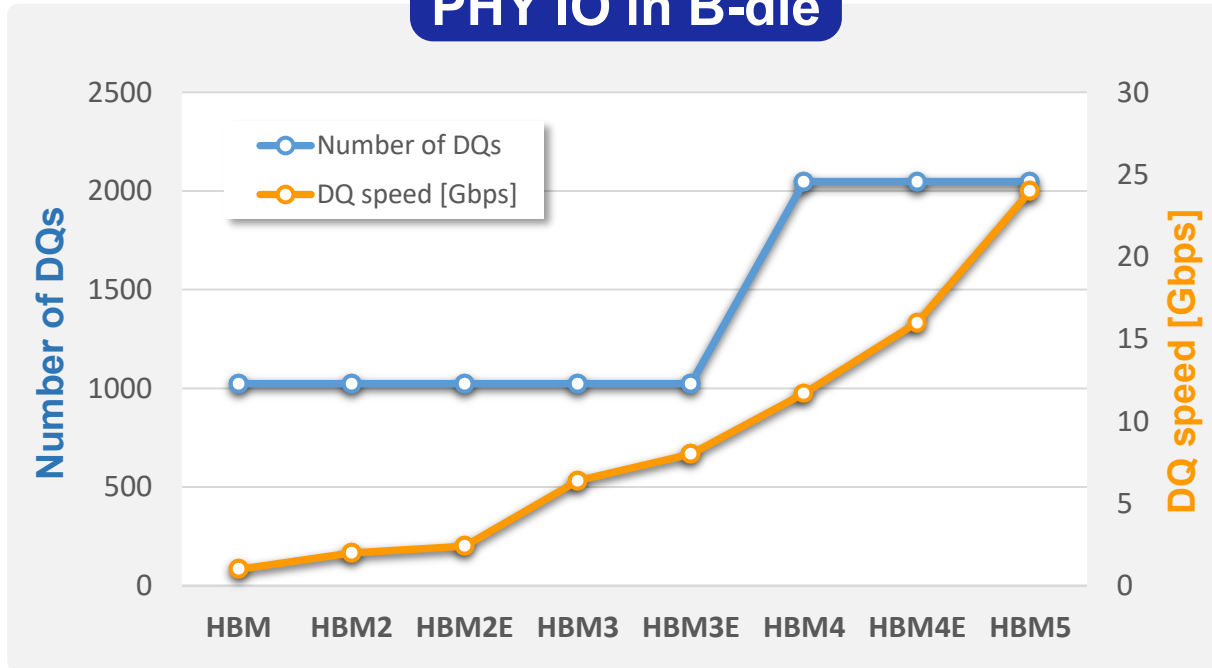
Key Factors Limiting BW Scaling

■ Primary bottlenecks:

- TSV counts & pitch limits (B-die and C-die stack)
- B-die PHY I/O count and speed limitations

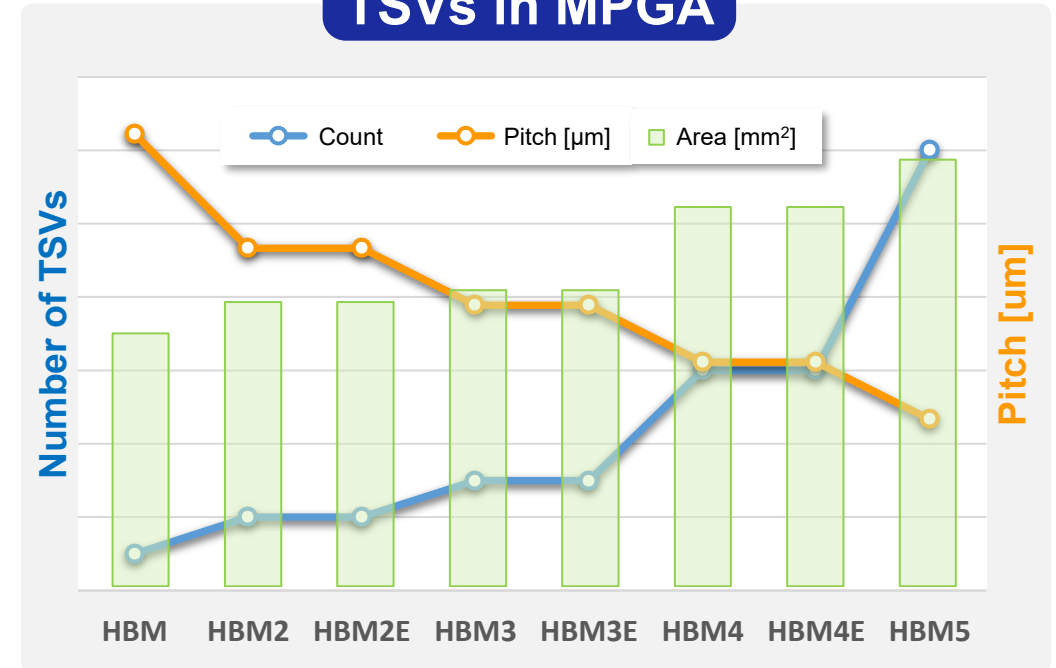


PHY IO in B-die



TSVs in MPGA

※ Micro-Pillar Grid Array

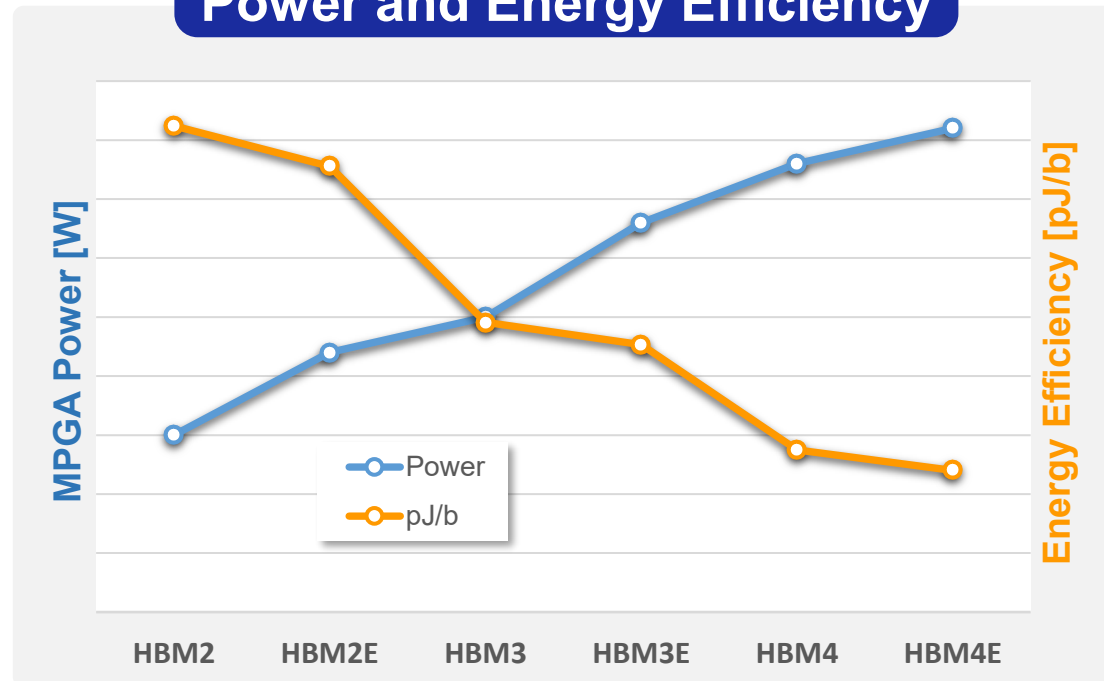


Process Node Trend

Year			2016	2019	2021	2023	2026	2027
Generation			HBM2	HBM2E	HBM3	HBM3E	HBM4	HBM4E
Process	HBM	C-die	2*nm	1*nm	1*nm	1*nm	1*nm	1*nm
		B-die	2*nm	1*nm	1*nm	1*nm	4nm	4nm
	xPU SoC		16n/12nm	8n/4nm	4nm	4nm	3nm	<3nm

DRAM process
 Logic process

Power and Energy Efficiency



■ Key observations:

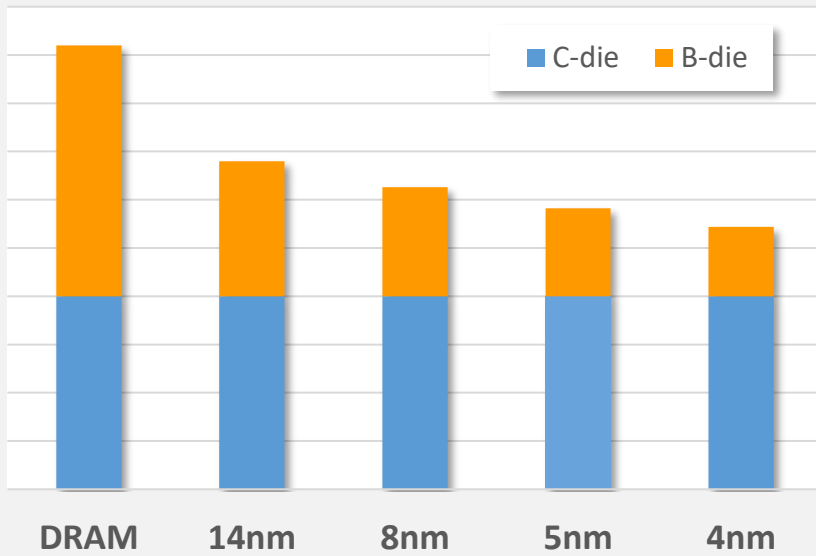
- MPGA power continues to rise despite improving energy efficiency.
- Starting from HBM4, adopting advanced logic processes in B-die design is essential to curb power growth.
- The process gap between B-die and xPU SoC is narrowing.

Benefits of Using Advanced Logic Processes

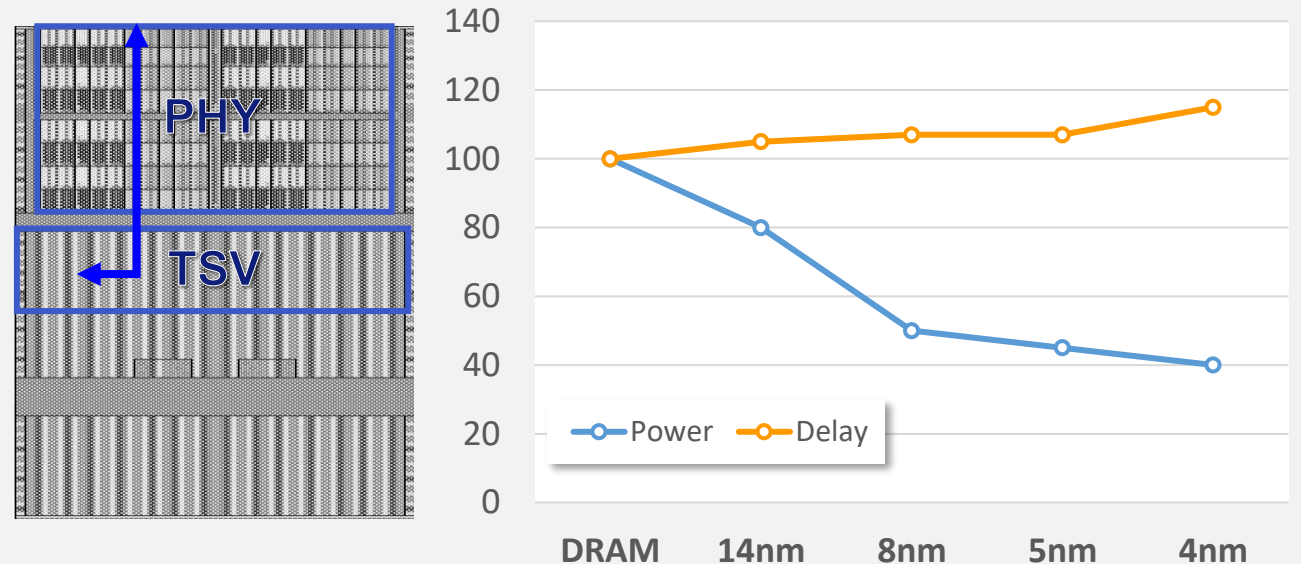
■ Samsung applied D1c and logic 4nm processes to HBM4.

- The primary objective is power reduction.
- Active area minimization is another key goal.
- Marks the beginning of true DRAM and advanced logic integration.

HBM4 B-die & C-die Power



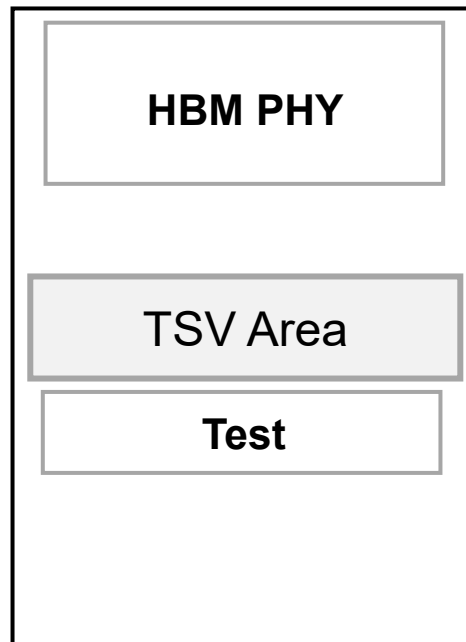
TSV-to-PHY Repeater Power & Delay



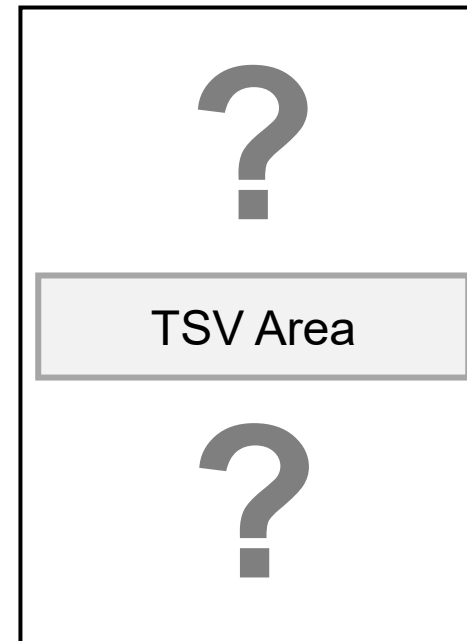
Toward Custom HBM: Concept and Motivation

■ Exploring value-added features for the logic B-die:

- Standard HBM (sHBM) B-die: limited to basic data & test path functions
- Key idea: leverage advanced logic processes to build SoC-like B-dies
- Custom HBM (cHBM): customizes B-die logic while sharing standard C-die stacks



B-die floorplan of sHBM



B-die floorplan of cHBM

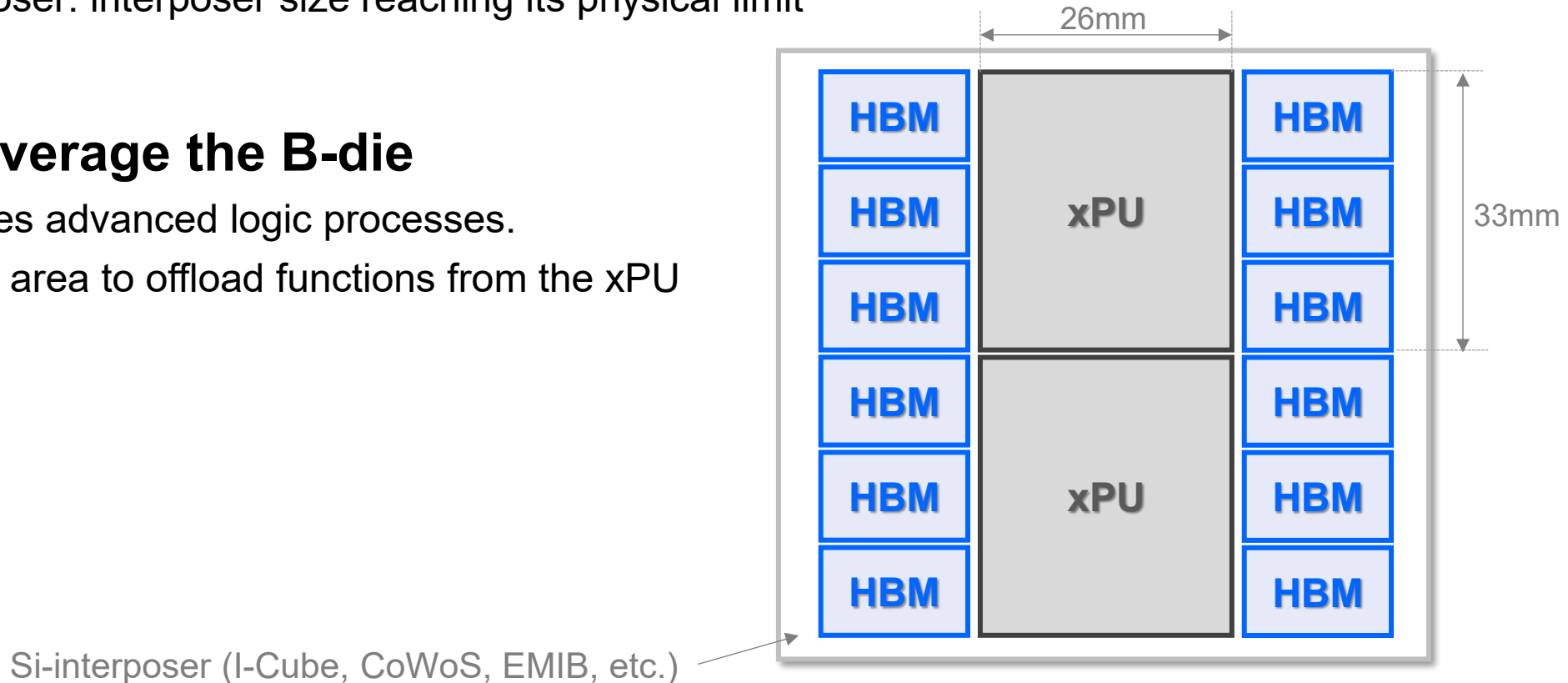
Phase #1: xPU Area Reclamation

■ Limits of traditional performance scaling

- Traditional approaches reaching physical limits
- Node shrink: process scaling is slowing down
- Larger monolithic die: reaching reticle size limits
- Multi-chiplet interposer: interposer size reaching its physical limit

■ The solution: leverage the B-die

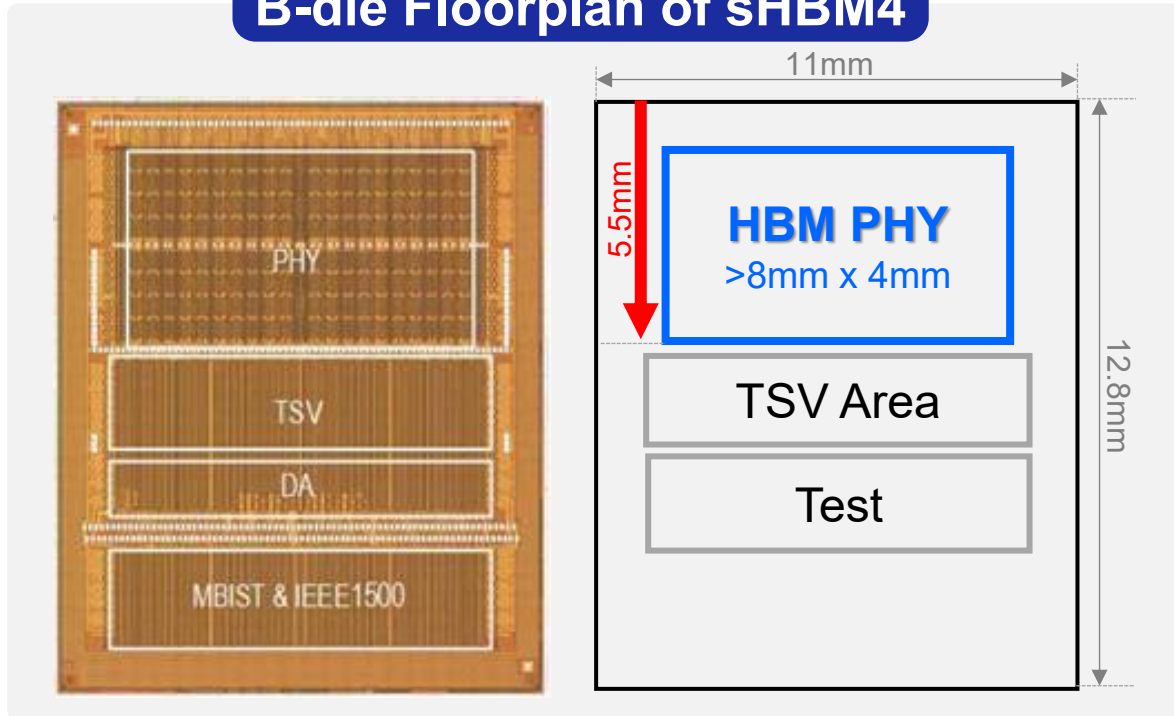
- B-die already utilizes advanced logic processes.
- Utilize ample B-die area to offload functions from the xPU



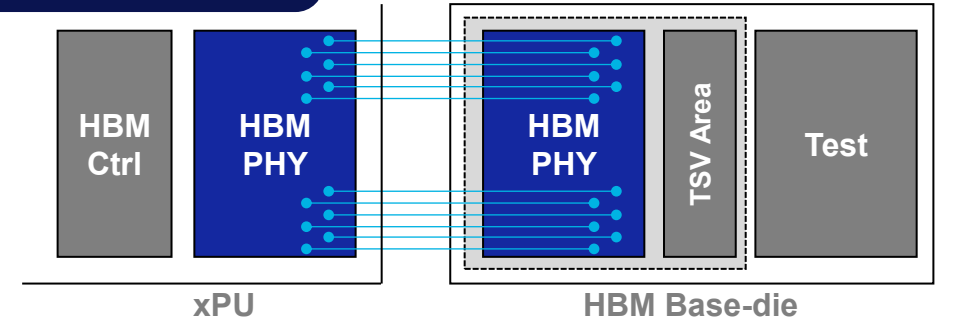
Phase #1: PHY Area Reduction

- **Conventional limits:** sHBM PHY is the largest block on the B-die.
 - cHBM solution replaces traditional HBM PHY with a D2D interface via advanced logic processes.
✕ Die-to-Die
- **Key benefits:**
 - Smaller footprint & shorter channel significantly improve energy efficiency (low pJ/b).
 - Area efficiency frees up valuable silicon area for xPU area expansion.

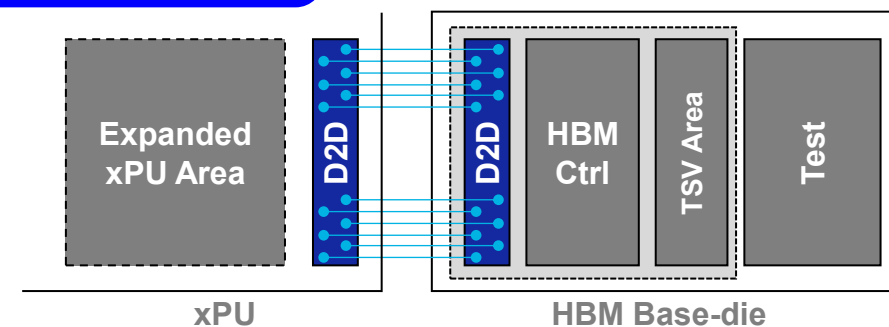
B-die Floorplan of sHBM4



Standard HBM



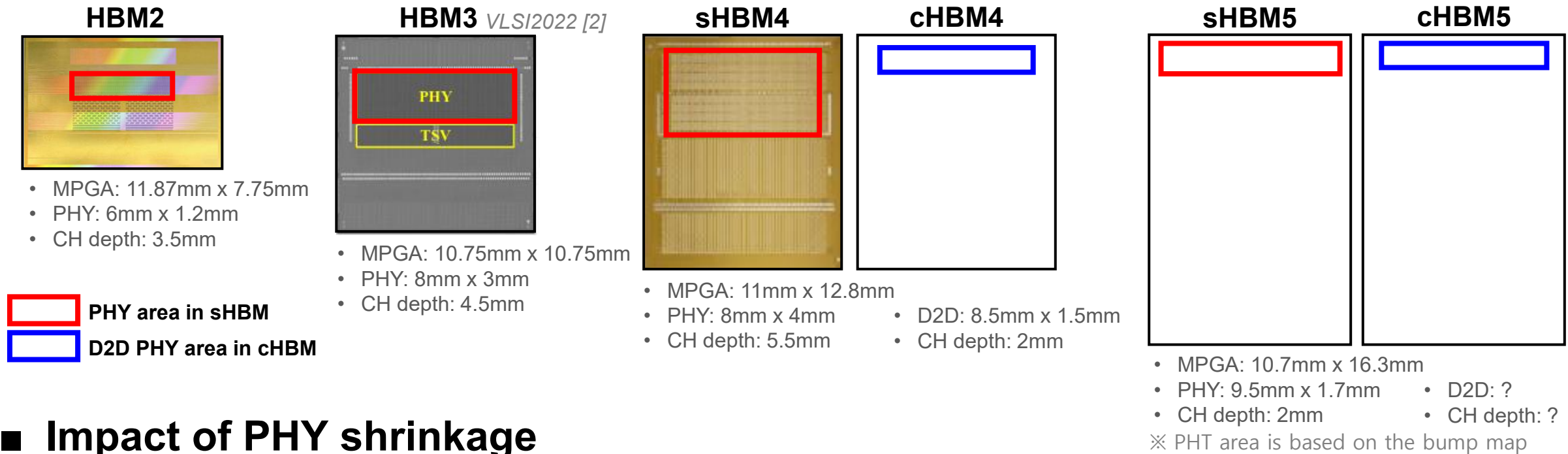
Custom HBM



Phase #1: Generational Area Reduction

■ PHY footprint scaling across HBM generations

- Conventional trend (HBM2→sHBM4): PHY footprint and CH length continuously expanded for higher BW.
- The inflection point (advanced logic): adopting advanced logic processes dramatically reduces PHY and D2D areas starting from HBM4 to HBM5.



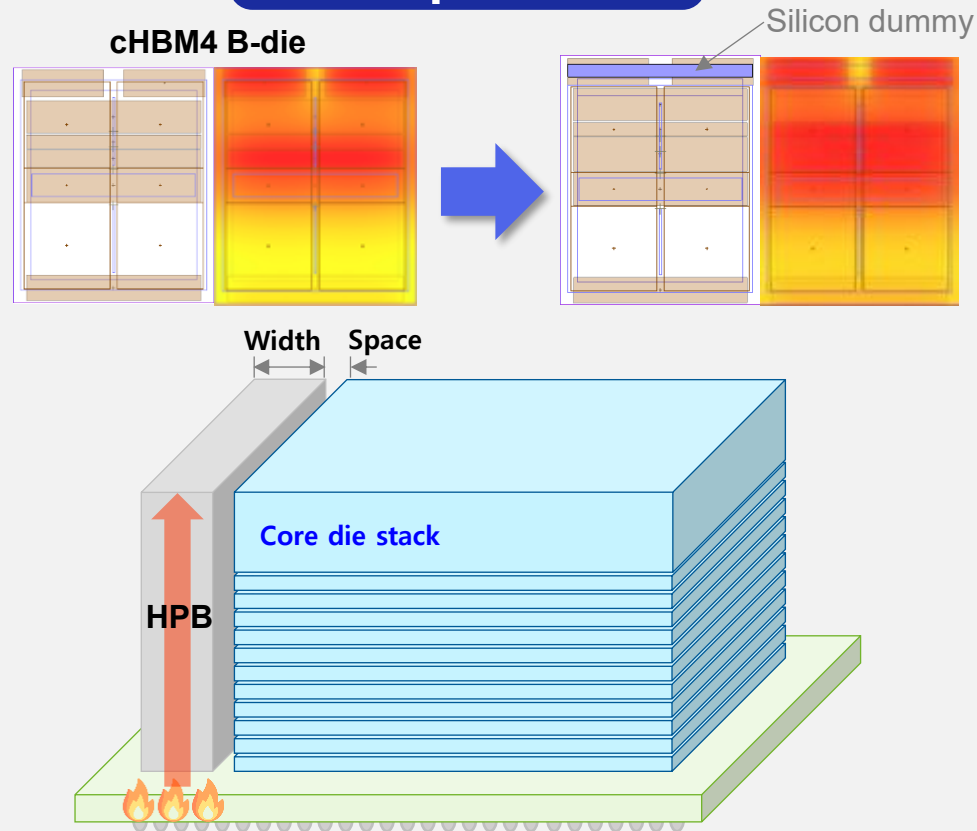
■ Impact of PHY shrinkage

- Benefit: shorter channel length significantly improves energy efficiency (lower pJ/b).
- Challenge: higher power density in smaller PHY areas creates thermal hotspots.

Phase #1: Thermal Challenge & Solution

- **HPB (Heat Path Block):** key thermal solution for high-power-density PHYs
 - Proven technology: built upon Samsung's cHBM4 design experience
 - Thermal impact: dramatically reduces peak temperature (>35%) with >50% PHY coverage ratio

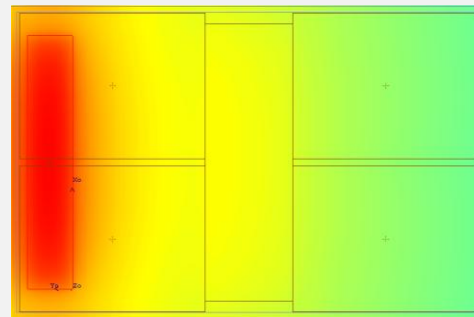
Concept of HPB



Effect of HPB

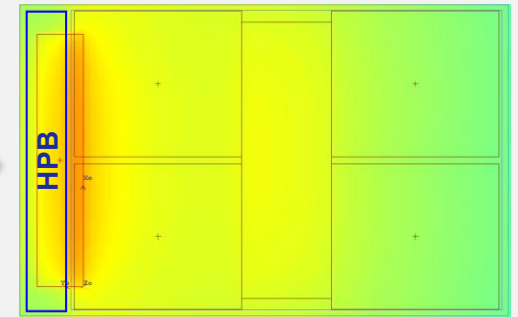
Product	sHBM4E	sHBM5
I/O Speed	14Gbps	>28Gbps (2x)
Power Density (W/mm ²)	0.5	>2.0
Thermal Status	●	● → ●

sHBM5 Without HPB



sHBM5 With HPB

(HPB coverage target: >50%)



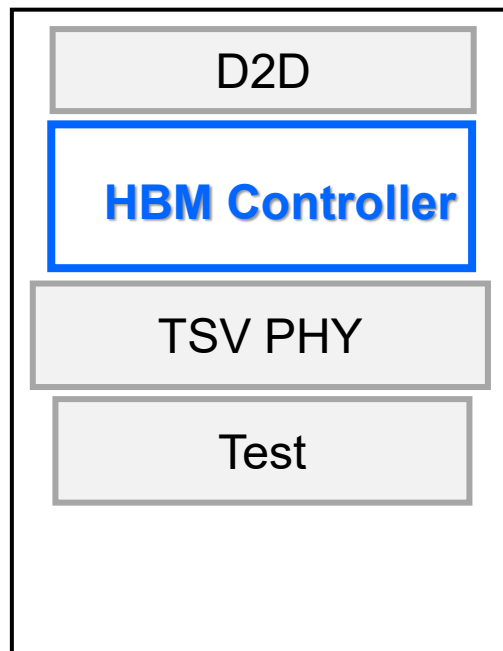
Phase #1: Memory Controller Offloading

■ Offloading xPU SoC logic to the cHBM B-die:

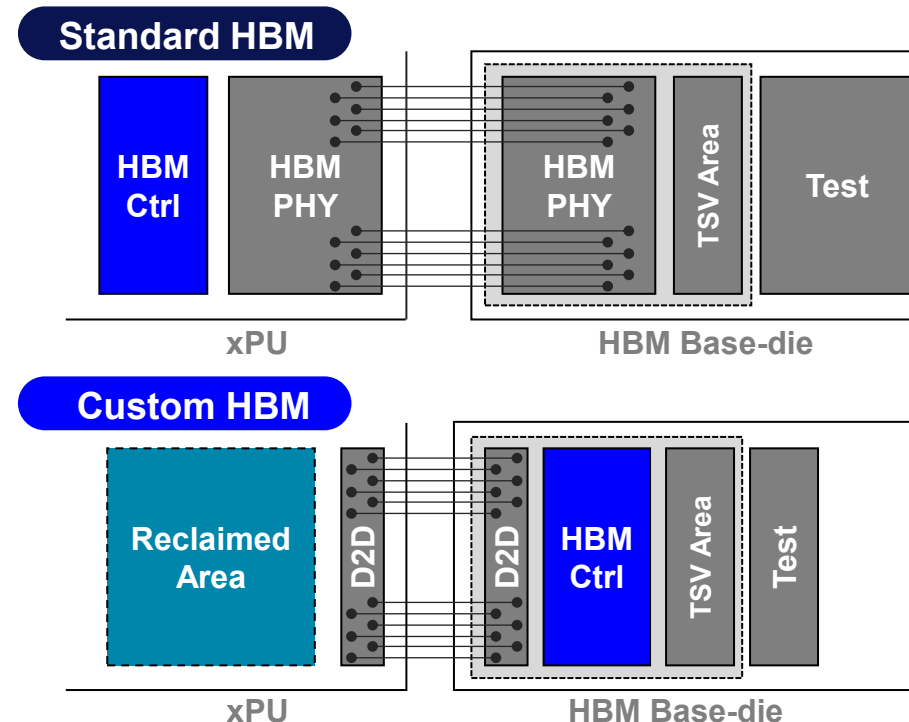
- Area efficiency: reclaims valuable xPU SoC silicon area
- Prime target: the memory controller(MC)

■ Feasibility:

- Thermal overhead: thermal impact from the offloaded MC is manageable.
- Current status: conventional HBM MCs are actively being ported into cHBM.



B-die floorplan of custom HBM

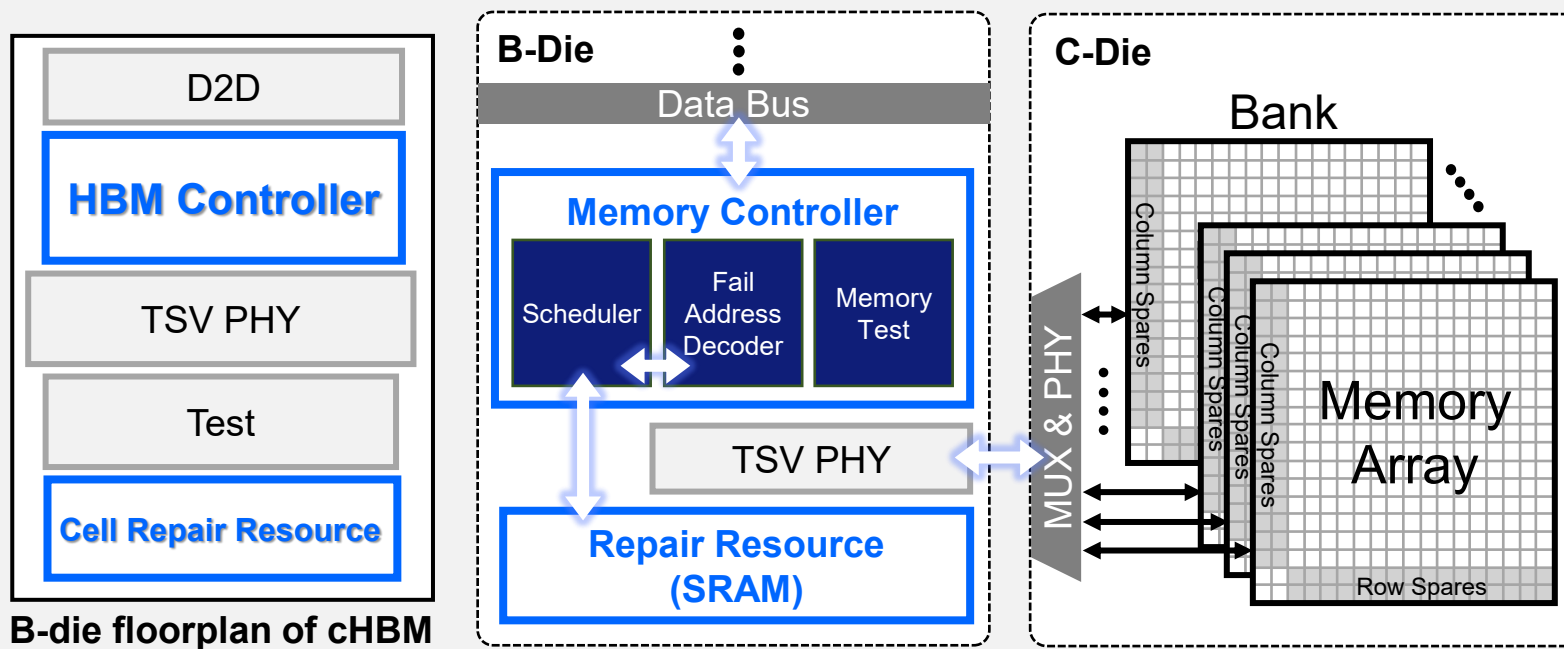


Phase #1: Near-MC: SRAM-Based Cell Repair Scheme

■ Enhanced RAS via B-die SRAM integration:

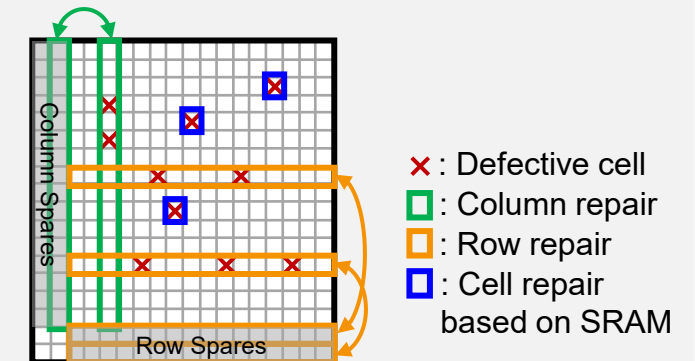
- Area efficiency: leverages unused B-die space for SRAM repair resources
- Fine-grained repair: enables cell-level fail-address decoding and redirection in the MC
- Resource sharing: shares repair capacity across channels

Concept of SRAM-Repair Scheme



Scheme Comparison

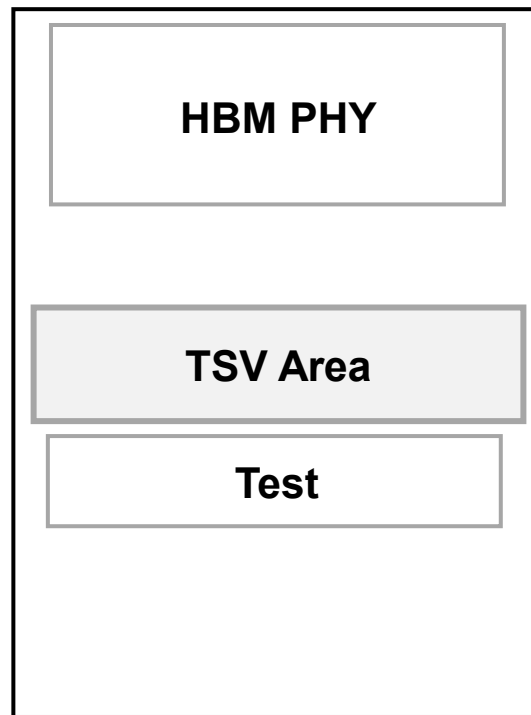
Conventional scheme	Proposed scheme
• Using repair resource in C-die • Limited amount of resource and limited flexibility	• Using repair resource in B-die SRAM • Large amount of resource and higher flexibility



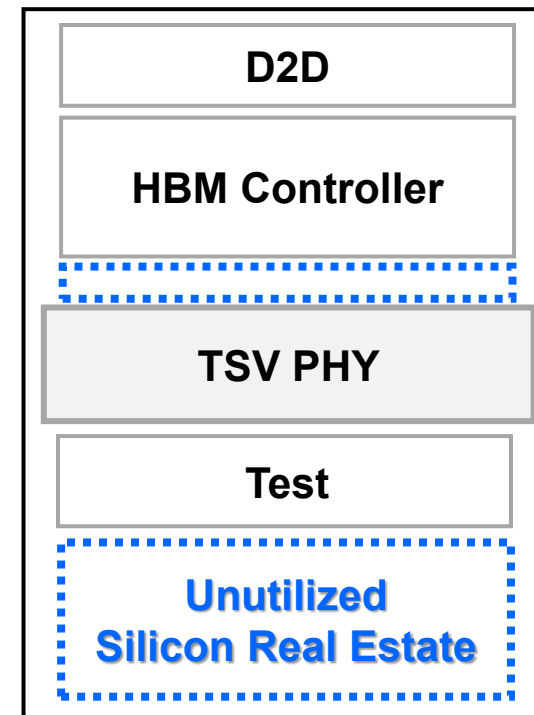
Phase #2: Additional Functions

■ Motivation for functional expansion

- Fixed footprint: the overall HBM footprint is primarily bounded by the C-die stack.
- sHBM limitation: most of the sHBM B-die is passively used for routing.
- New opportunity: even with MC offloading, substantial unutilized area remains on the B-die – enabling extra functionality in Phase #2.



B-die floorplan of sHBM



B-die floorplan of cHBM

Phase #2: RAS and Test

※ Reliability + Availability + Serviceability

■ SoC-level RAS capability:

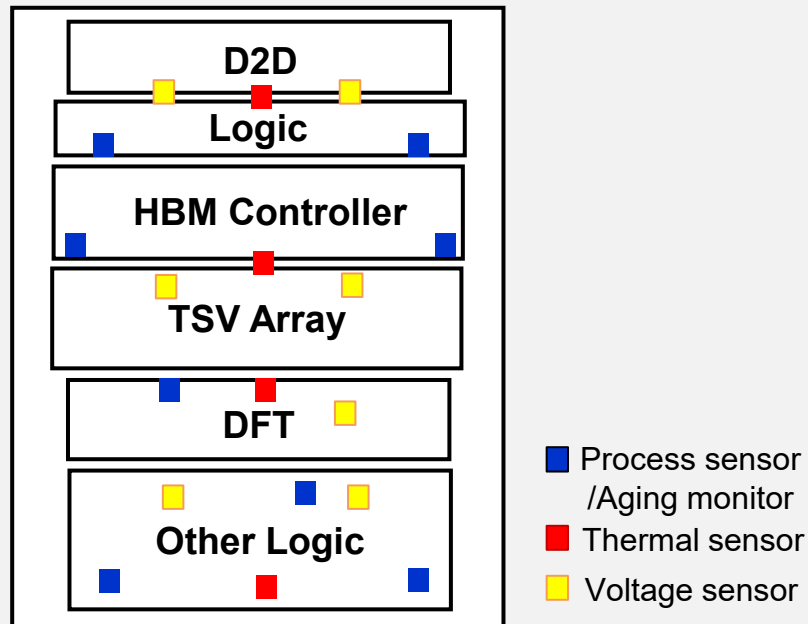
- Unlike sHBM, cHBM natively integrates advanced RAS capabilities into the B-die.
- Thermal, voltage, process, and aging sensors enable real-time, high-speed telemetry.

■ Advanced testing & yield improvement

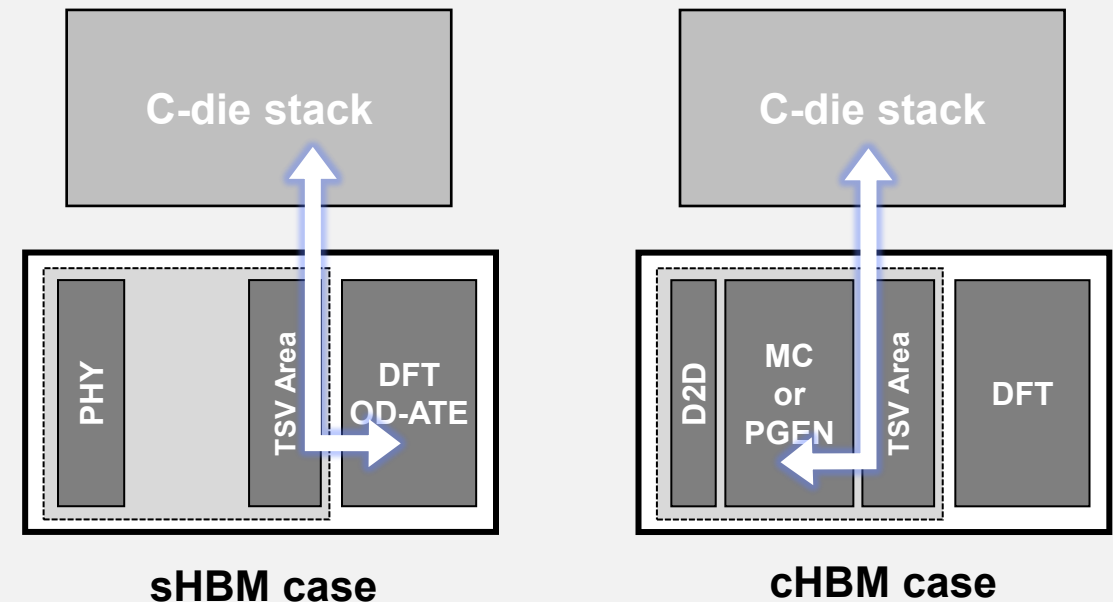
- On-chip self-test schemes(OD-ATE/PGEN) improve test coverage and manufacturing yield.

※ On-die ATE/Pattern Generator

Example of Sensor Deployment



MPGA Self-Test Scheme

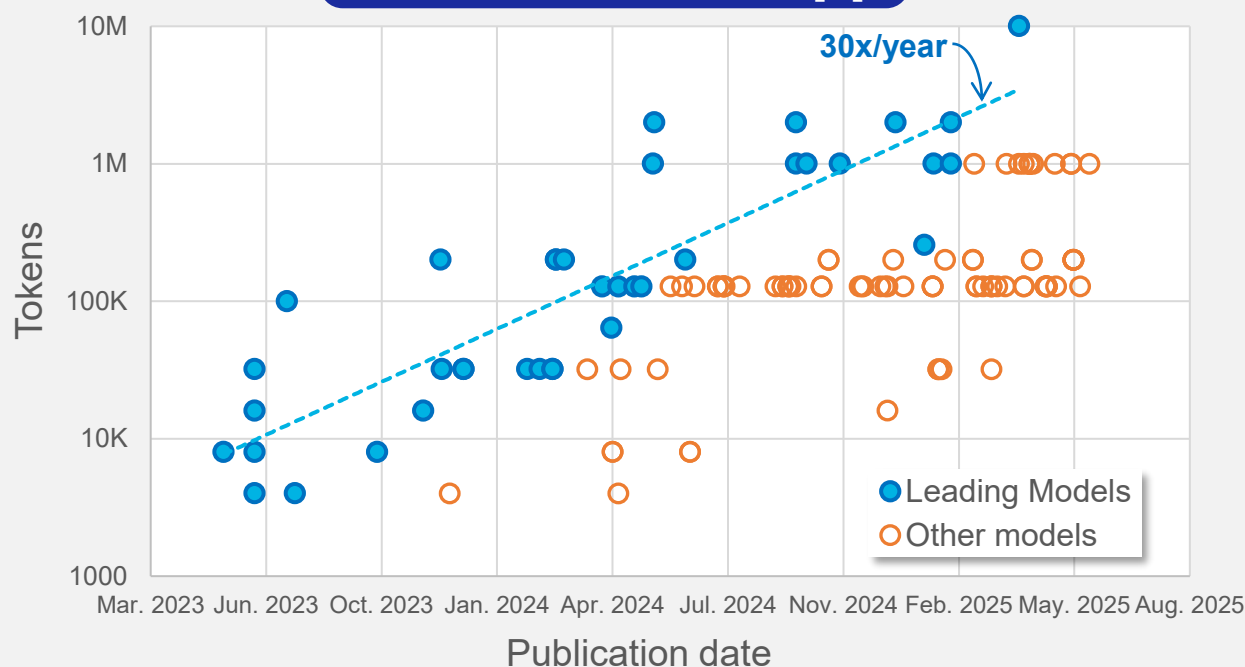


Phase #2: Need for Capacity

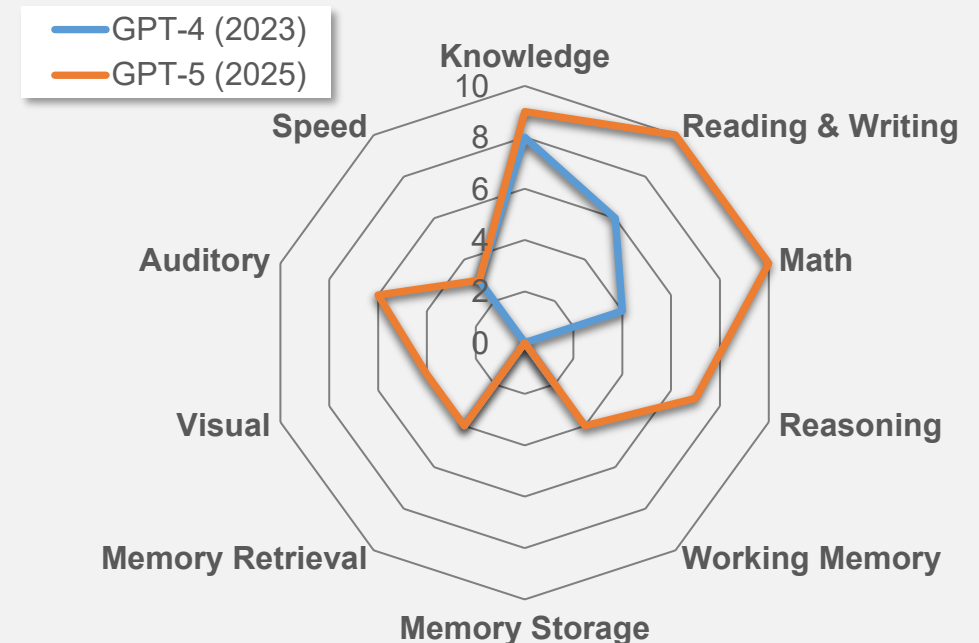
■ Growing demand for memory capacity

- Explosive context window growth: rapidly expanding context lengths (30x/year) drive massive KV cache memory requirements.
- AGI scaling bottleneck: long-term memory storage and retrieval remain the most critical bottlenecks for next-gen AI models.
- Key takeaway: next-gen AI SoC architecture urgently demands higher memory capacity alongside BW.

Context Window [3]



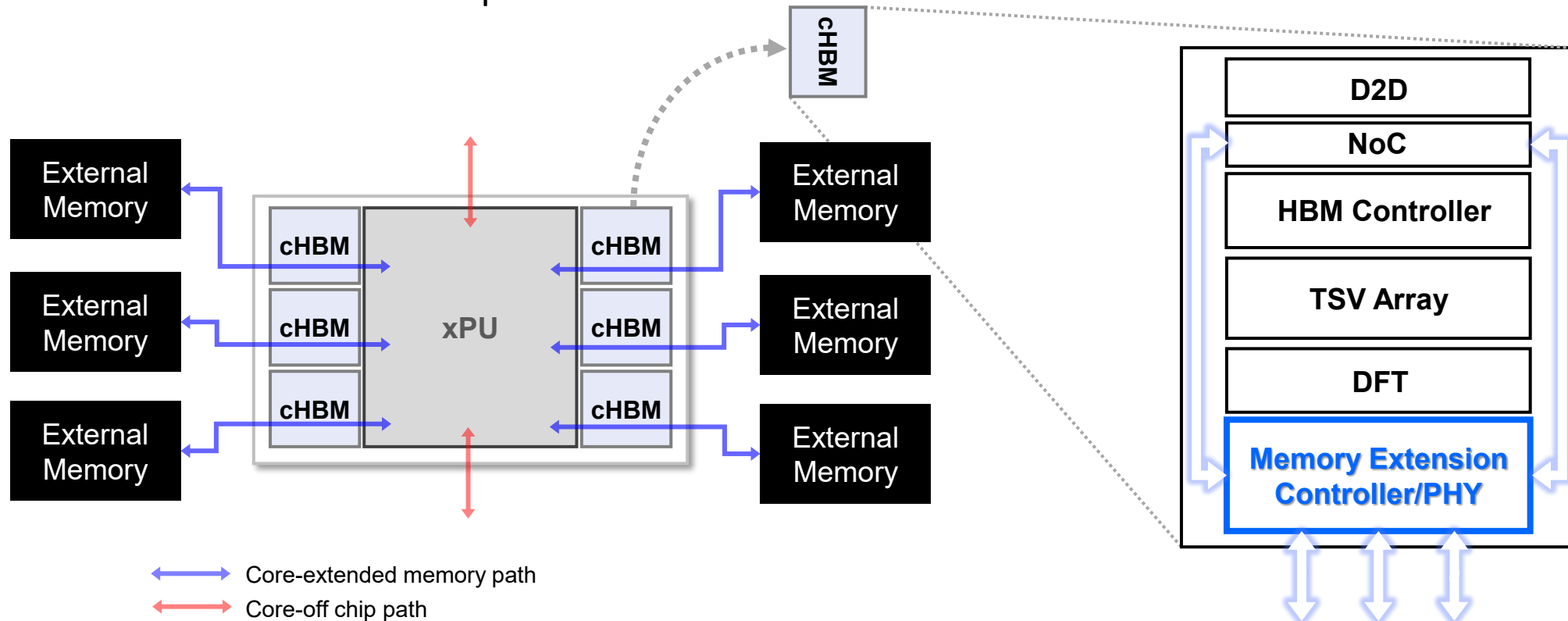
AGI Score Summary [4]



Phase #2: Memory Extension

■ Direct memory extension via B-die

- Outer shoreline expansion: utilizes the outer B-die shoreline to connect external memory directly.
- Dedicated IP integration: houses dedicated PHYs and controllers in the B-die to manage off-chip memory expansion.
- Performance advantage: delivers significantly higher bandwidth and lower latency compared to conventional PCIe-based expansion.



Phase #2: Processing Elements Offloading

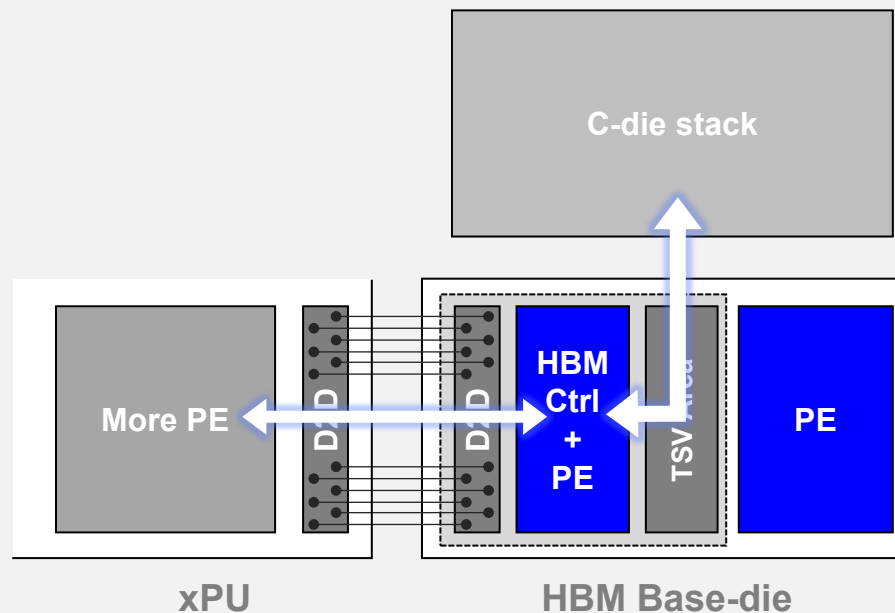
■ Concept & benefit:

- Offloads partial xPU SoC computation to Processing Elements (PEs) in the B-die.
- Reduces D2D bandwidth requirements, lowering power and thermal overhead.

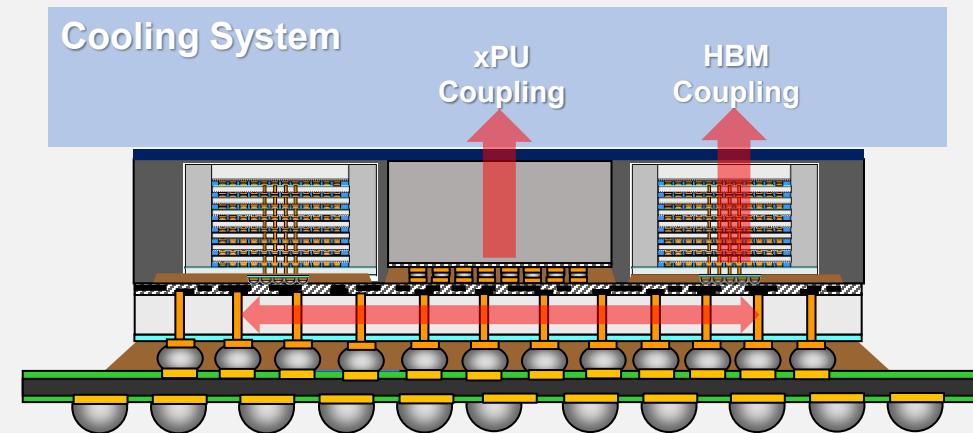
■ Key challenges:

- Limited silicon area and increased thermal density from B-die PEs.

PE in B-die Configuration



AI SoC Cooling



Phase #2: 2.5D Integration & aHBM

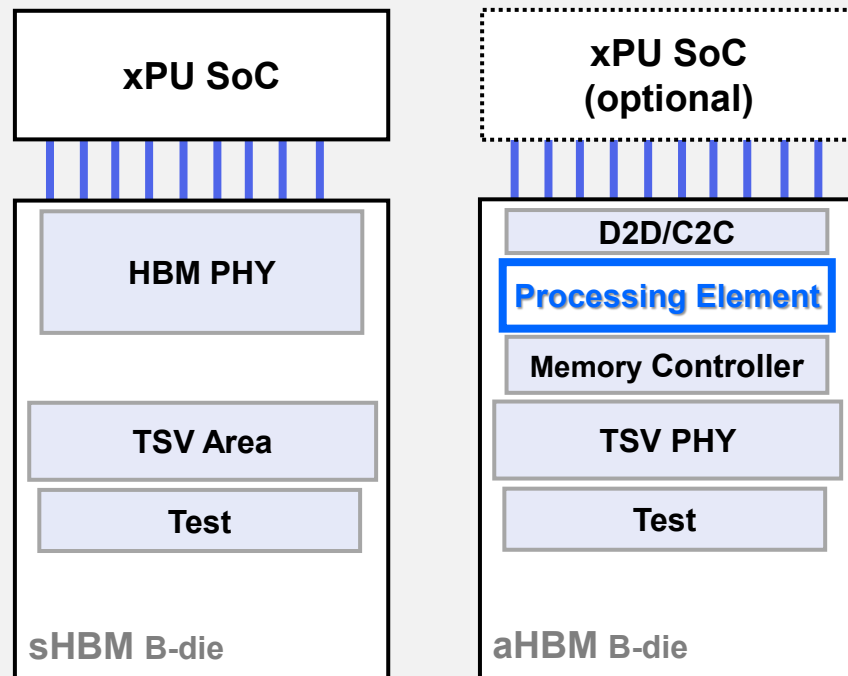
■ Advanced HBM (aHBM) in 2.5D systems:

- Offloads xPU computing to PEs integrated into the B-die.

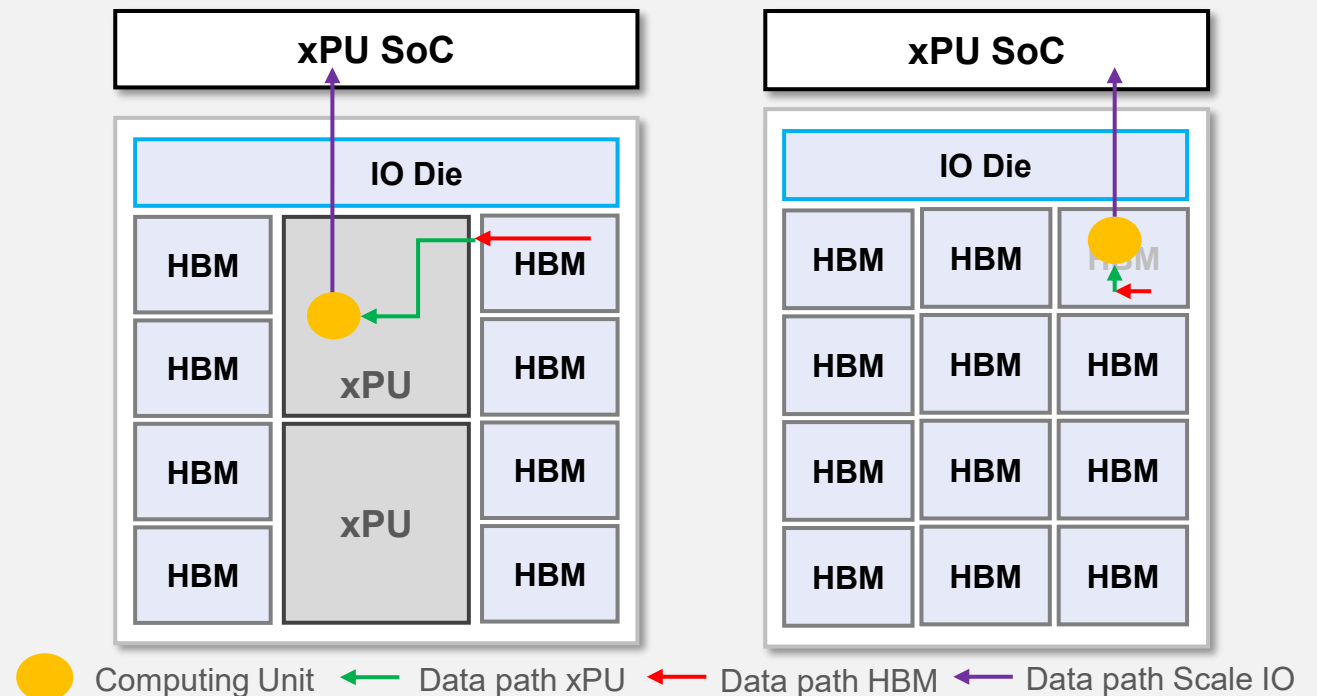
■ System-level benefits:

- Data path optimization: dramatically minimizes data movement across the interposer.
- Maximized efficiency: delivers breakthrough improvements in overall system power efficiency.

B-die Configuration



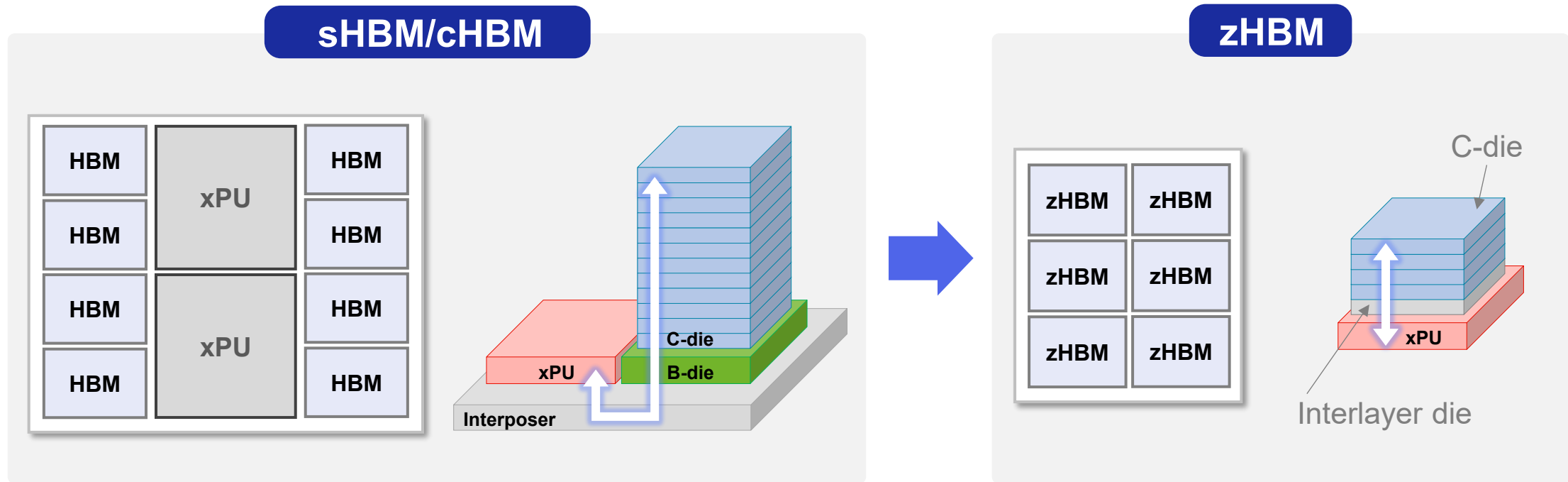
System Comparison



Phase #3: 3D Integration

■ Vertical integration and zHBM

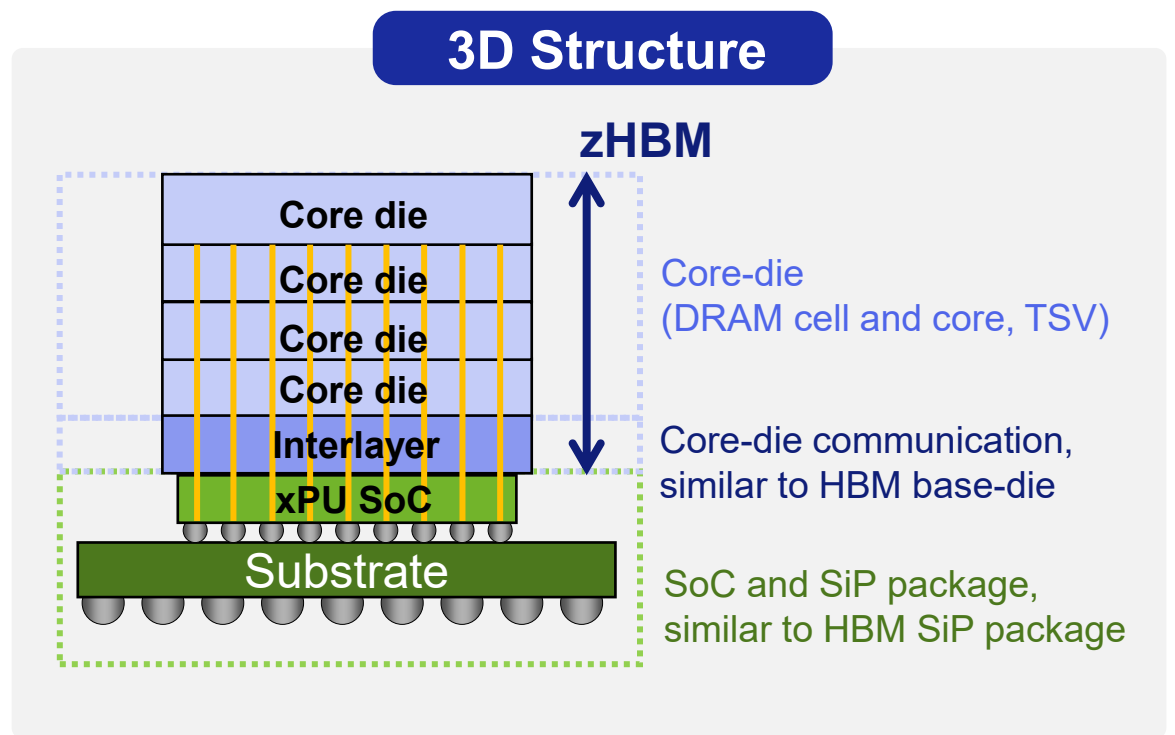
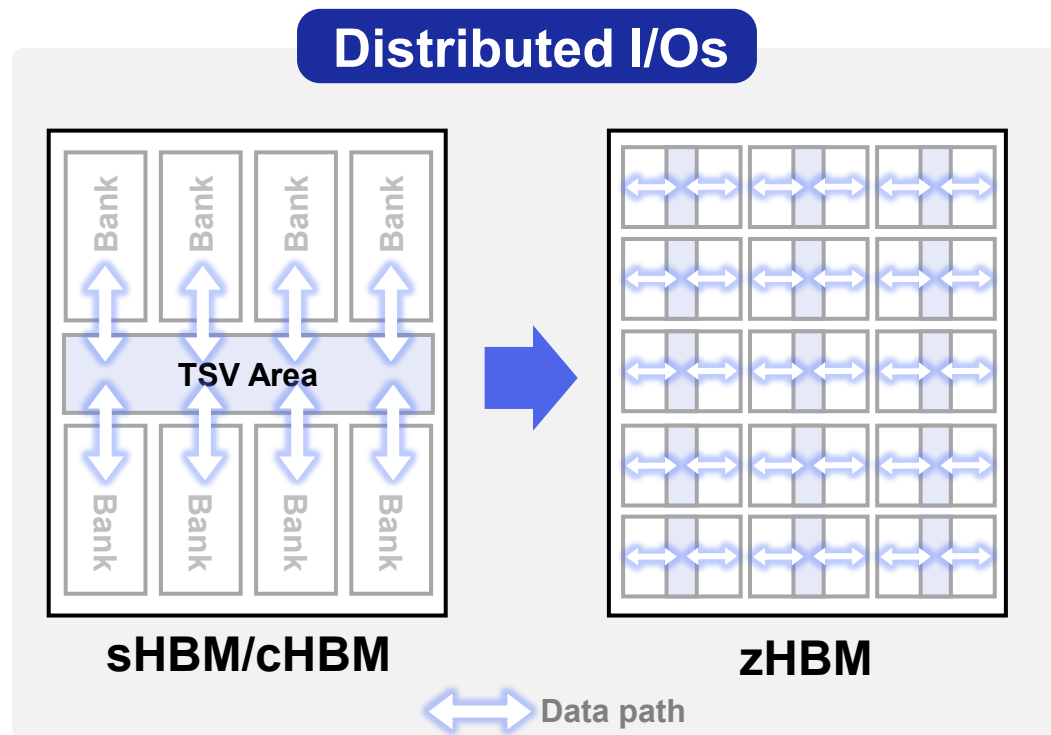
- Industry trend: AI memory architecture is rapidly moving toward tightly coupled solutions.
- The challenge (AGI & inference): maximizing bandwidth (TPS) within strict power limits is critical.
× Tokens Per Second
- The ultimate solution (zHBM): true 3D vertical integration of the xPU and C-die stack, eliminating the 2.5D interposer.



Phase #3: Concept of zHBM

- **Key differentiators:** distributed I/Os and a true 3D structure

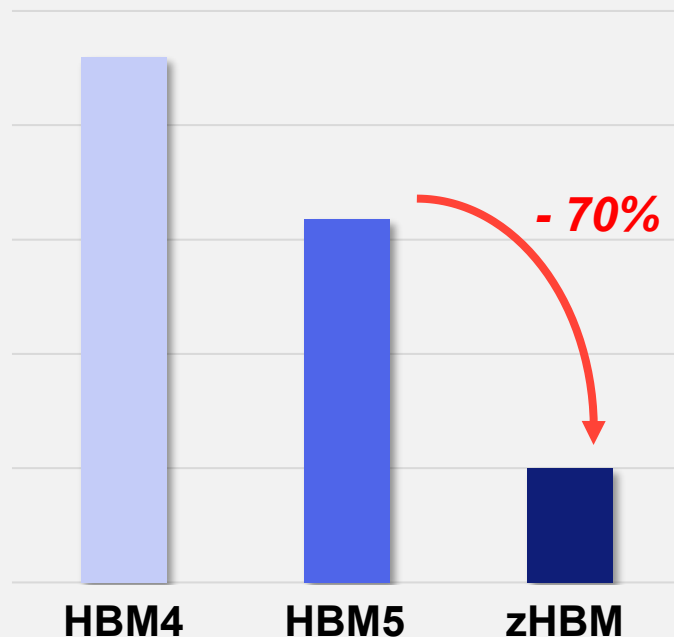
- Distributed I/Os minimize data travel distance within the HBM stack.
- 3D structure eliminates conventional 2D interfaces (e.g., HBM PHY or D2D).
- Extreme efficiency targets ultra-low power consumption.



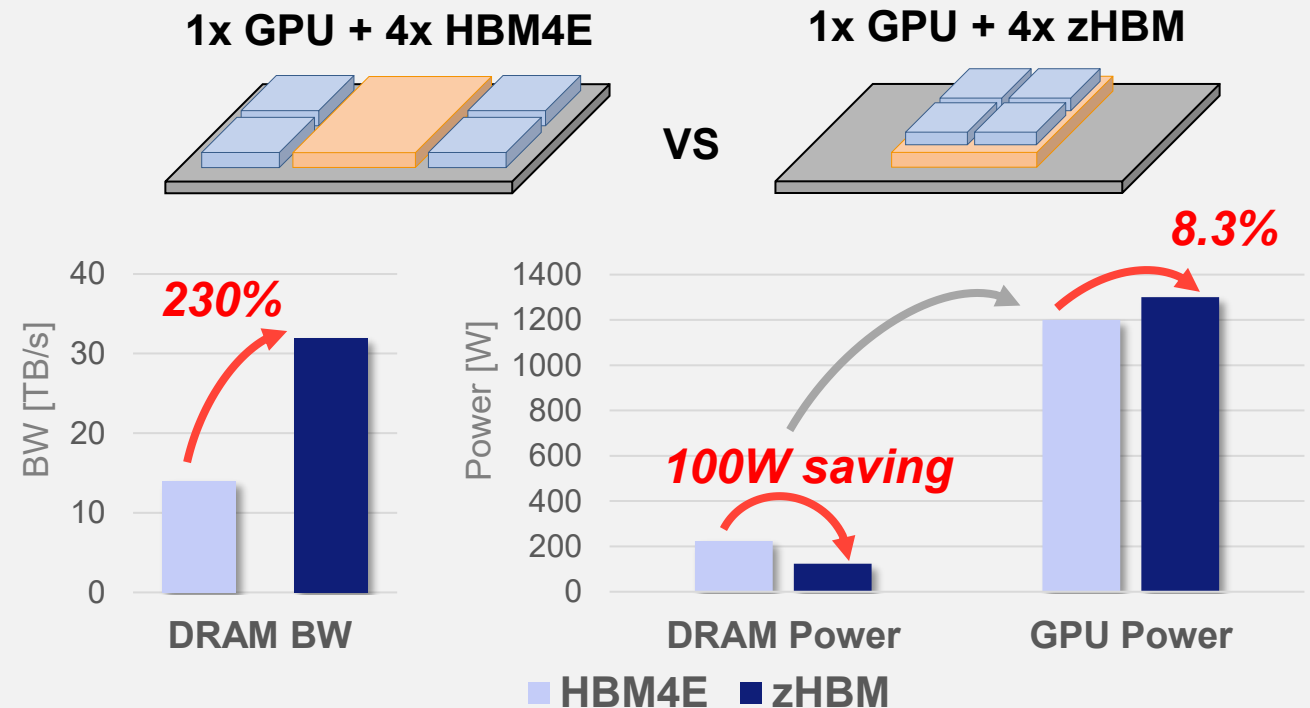
Phase #3: Advantages of zHBM

- **Power reduction:** the primary advantage is a dramatic decrease in I/O power.
 - I/O optimization removes SERDES (Data align/DQ I/O) to eliminate unnecessary power overhead.
 - System-level impact: this breakthrough in power efficiency translates directly into enhanced compute performance and thermal headroom.

Power Efficiency



System Level Benefit

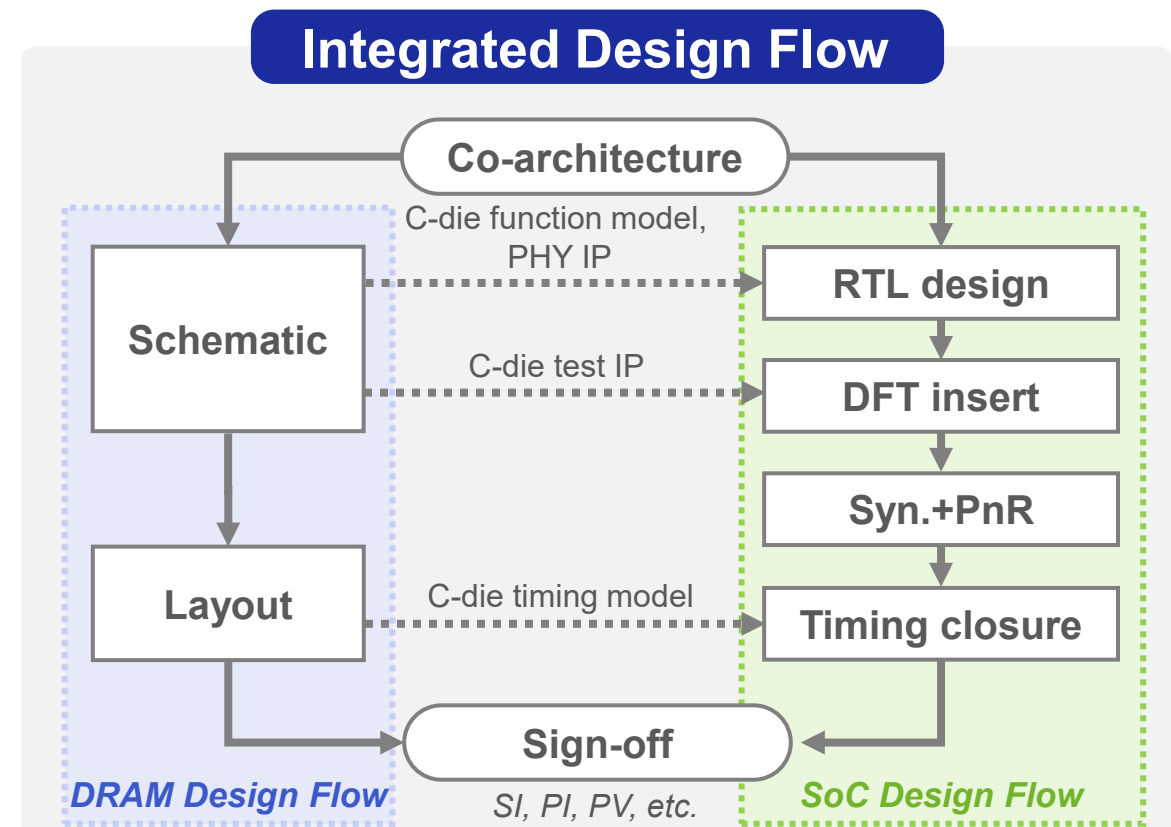
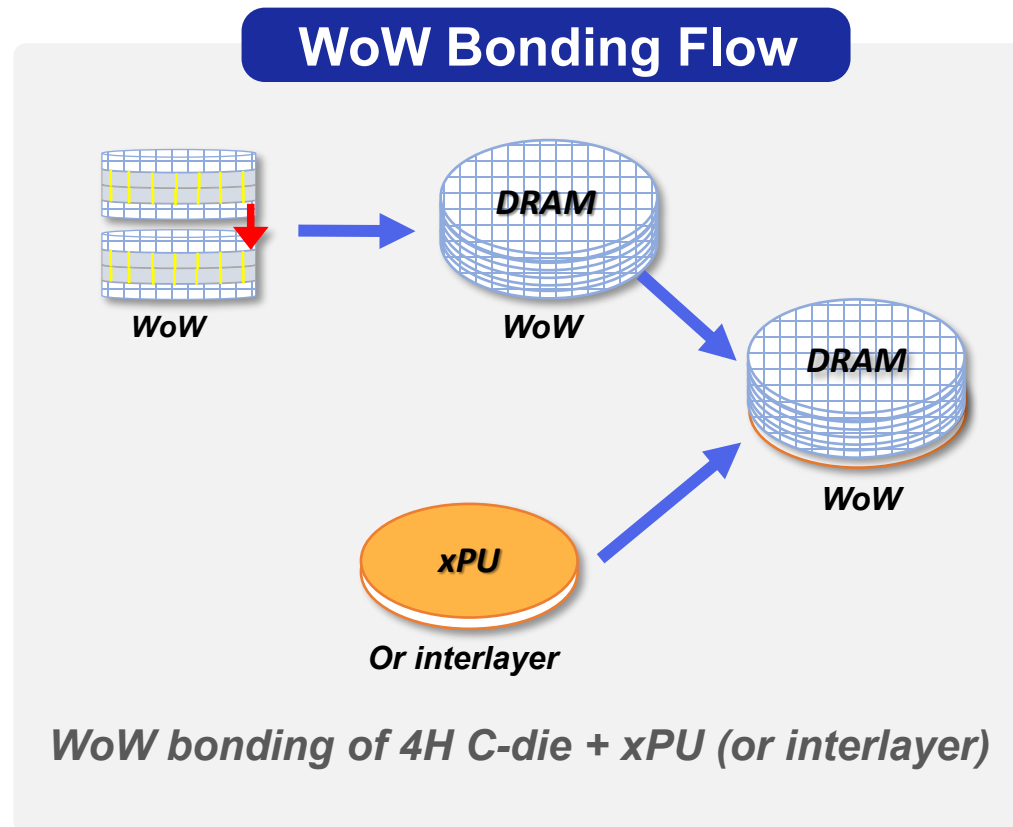


※ Assumption : GPU Power 1200W, 4 HBMs in SIP

Phase #3: Key Technologies of zHBM

■ Two categories of key technology

- Advanced packaging: WoW and HCB enable ultra-high I/O density.
※ Wafer-on-Wafer ※ Hybrid Copper Bonding
- Integrated methodology: A unified SoC-DRAM design and verification flow is essential for successful co-architecture.



Summary and Conclusions

■ Emergence of cHBM:

- Rapidly evolving AI xPU SoCs demand tailored memory architectures.
- Applying advanced logic processes to the HBM B-die unlocks unprecedented architectural flexibility.

■ Three-step evolutionary roadmap:

- Phase #1 (offloading & optimization): reclaims xPU silicon area by adopting compact D2D PHYs and offloading memory controllers to the B-die.
- Phase #2 (functional expansion): integrates value-added logic—such as telemetry sensors, external memory expansion, and PEs—directly into the B-die.
- Phase #3 (true 3D convergence): eliminates 2D/2.5D interfaces via zHBM, achieving direct vertical integration of xPU logic and DRAM stacks for maximum energy efficiency.

Thank You.

References

- [1] **AI Revolution Driven by Memory Technology Innovation**, ISSCC, 2025
- [2] **A 16 GB 1024 GB/s HBM3 DRAM with On-Die Error Control Scheme for Enhanced RAS Features**, VLSI Symposium, 2022
- [3] **LLMs now accept longer inputs, and the best models can use them more effectively** <https://epoch.ai/data-insights/context-windows>
- [4] **A Definition of AGI** <https://arxiv.org/pdf/2510.18212>